Laser Thresholds and Symmetry-Breaking Pitchforks

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Notation: Laser variables and parameters

Let

$n(t) \geq 0$	number of photons
$N(t) \geq 0$	number of excited atoms
$G > 0$	gain coefficient
$k > 0$	photon-loss rate
$N_0 > 0$	excited-atom level without laser action
$\alpha > 0$	depletion coefficient

Idea: Photon gain and loss

Stimulated emission contributes GnN, while photon loss contributes $-kn$:

$$\dot{n} = GnN - kn.$$

Idea: Excited-atom depletion

Approximate the number of excited atoms by

$$N = N_0 - \alpha n,$$

so a larger photon population leaves fewer excited atoms available. Because $n \geq 0$ and $N \geq 0$, the physically admissible photon states satisfy

$$0 \leq n \leq \frac{N_0}{\alpha}.$$

Question: Where is the laser threshold?

Reduce the photon balance to a scalar equation, find all equilibrium branches, and classify the physically admissible states.

Solution. Substitution of $N = N_0 - \alpha n$ gives

$$\dot{n} = Gn(N_0 - \alpha n) - kn = (GN_0 - k)n - \alpha G n^2.$$

Define

$$\mu = GN_0 - k.$$

Then

$$\dot{n} = f(n;\mu), \qquad f(n;\mu) = n(\mu - \alpha G n).$$

At the endpoints of the physical interval,

$$f(0;\mu) = 0, \qquad f\left(\frac{N_0}{\alpha};\mu\right) = -\frac{kN_0}{\alpha} < 0.$$

The reduced vector field is smooth, so uniqueness prevents a solution from crossing through $n = 0$, while the vector field at N_0/α points into the interval. Therefore every solution that starts in $[0, N_0/\alpha]$ remains there for all forward times for which it is defined. Because this interval is compact and the vector field is continuously differentiable, a solution confined to it cannot become unbounded in finite time. Hence every such solution is defined for all $t \geq 0$.
The mathematical equilibrium branches are

$$n_0^* = 0, \qquad n_1^* = \frac{\mu}{\alpha G}.$$

The physical parameters imply $\mu > -k$. Within this admissible parameter range, the branch n_1^* is defined for every μ, but it is a physically admissible photon state only when $\mu \geq 0$. Since

$$f_n(n;\mu) = \mu - 2\alpha G n,$$

their rates are

$$f_n(0;\mu) = \mu, \qquad f_n\left(\frac{\mu}{\alpha G};\mu\right) = -\mu.$$

For $\mu = 0$, the equation on the physical state space is

$$\dot{n} = -\alpha G n^2,$$

and its solution is

$$n(t) = \frac{n_0}{1 + \alpha G n_0 t}.$$

Thus every $n_0 > 0$ converges to 0, while $0 \leq n(t) \leq n_0$ for every $t \geq 0$. Hence $n^* = 0$ is both attracting and Lyapunov stable at the nonhyperbolic threshold. Combining this one-sided argument with the derivative test away from $\mu = 0$, the branches exchange stability at $\mu = 0$. On the physical state space $0 \leq n \leq N_0/\alpha$,

$$\begin{array}{r|c}
N_0 < k/G & n^* = 0 \text{ is asymptotically stable} \\
N_0 = k/G & n^* = 0 \text{ is nonhyperbolic and asymptotically stable} \\
N_0 > k/G & \begin{array}{l} n^* = 0 \text{ is unstable,} \\ n^* = (GN_0 - k)/(\alpha G) > 0 \text{ is asymptotically stable.} \end{array}
\end{array}$$

The pump threshold is

$$\boxed{N_{0,c} = \frac{k}{G}}.$$

Definition: Reflection symmetry

Let $I \subseteq \mathbb{R}$ be symmetric about 0, let $R \subseteq \mathbb{R}$ be a parameter set, and let $f : I \times R \to \mathbb{R}$. The scalar vector field has **reflection symmetry** if

$$f(-x; r) = -f(x; r) \qquad \text{for every } x \in I \text{ and } r \in R.$$

Observation: Consequences of reflection symmetry

If $f(-x; r) = -f(x; r)$, then $x = 0$ is an equilibrium and every nonzero equilibrium occurs with its reflected partner. If f is five times continuously differentiable in x near $x = 0$, then, for each fixed r,

$$f(x; r) = a_1(r)x + a_3(r)x^3 + a_5(r)x^5 + o(x^5) \qquad (x \to 0),$$

where

$$a_1(r) = f_x(0; r), \qquad a_3(r) = \frac{f_{xxx}(0; r)}{3!}, \qquad a_5(r) = \frac{f_{xxxxx}(0; r)}{5!}.$$

Definition: Pitchfork bifurcation

A **pitchfork bifurcation** occurs when a persistent equilibrium branch changes stability while a reflected pair of nonzero equilibrium branches meets it.

Definition: Supercritical pitchfork

Orient the parameter so that the origin changes from asymptotically stable for $r < 0$ to unstable for $r > 0$. The pitchfork is **supercritical** if asymptotically stable nonzero branches exist for $r > 0$.

Definition: Subcritical pitchfork

With the same parameter orientation, the pitchfork is **subcritical** if unstable nonzero branches exist for $r < 0$.

Question: How does the supercritical normal form break symmetry?

Analyze

$$\dot{x} = rx - x^3 = x(r - x^2).$$

Solution. The origin is an equilibrium for every r. For $r > 0$, there are also two equilibria

$$x^*_\pm = \pm\sqrt{r}.$$

Since

$$f_x(x; r) = r - 3x^2,$$

we have

$$f_x(0; r) = r, \qquad f_x(\pm\sqrt{r}; r) = -2r.$$

Therefore

r	$x^* = 0$	$x^* = \pm\sqrt{r}$
$r < 0$	asymptotically stable	absent
$r = 0$	asymptotically stable with algebraic decay	coincident at 0
$r > 0$	unstable	asymptotically stable

Question: How does the supercritical normal form break symmetry? (continued)

At $r = 0$, the solution of $\dot{x} = -x^3$ is

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}},$$

so convergence is algebraic.

Visualization: Supercritical potential landscape

For $\dot{x} = rx - x^3$, a potential is

$$V(x; r) = -\frac{r}{2}x^2 + \frac{1}{4}x^4.$$

r	critical points of V	landscape
$r < 0$	$x = 0$	one quadratic well
$r = 0$	$x = 0$	one quartic well
$r > 0$	$x = 0,\ \pm\sqrt{r}$	barrier at 0, wells at $\pm\sqrt{r}$

Question: Why does the hyperbolic-tangent model have a pitchfork?

For $\beta > 0$, analyze

$$\dot{x} = -x + \beta \tanh x,$$

and locate and classify its bifurcation.

Solution. The equation is reflection symmetric, and the origin is always an equilibrium. Since

$$\tanh x = x - \frac{x^3}{3} + O(x^5) \qquad (x \to 0),$$

we have

$$\dot{x} = (\beta - 1)x - \frac{\beta}{3}x^3 + O(x^5) \qquad (x \to 0).$$

The origin has rate $\beta - 1$. It is asymptotically stable for $0 < \beta < 1$, nonhyperbolic and asymptotically stable for $\beta = 1$, and unstable for $\beta > 1$.
For a nonzero equilibrium,

$$x^* = \beta \tanh x^*, \qquad \beta = \frac{x^*}{\tanh x^*}.$$

For $x > 0$,

$$\frac{d}{dx}\left(\frac{x}{\tanh x}\right) = \frac{\sinh x \cosh x - x}{\sinh^2 x} > 0,$$

and $x/\tanh x$ increases from 1 to ∞. Hence no nonzero equilibria exist for $0 < \beta \le 1$, while exactly one reflected pair exists for every $\beta > 1$. Along this branch,

$$f_x(x^*; \beta) = -1 + \frac{\beta}{\cosh^2 x^*} = -1 + \frac{2x^*}{\sinh(2x^*)} < 0$$

for $x^* \neq 0$, because $|\sinh y| > |y|$ for $y \neq 0$. Thus both nonzero branches are asymptotically stable, and the bifurcation at $\beta = 1$ is supercritical.

Question: What does the local subcritical normal form predict?

Analyze

$$\dot{x} = rx + x^3.$$

Solution. The origin has rate r. Nonzero equilibria satisfy

$$x^2 = -r,$$

so they exist for $r < 0$ and are

$$x^*_\pm = \pm\sqrt{-r}.$$

At either nonzero equilibrium,

$$f_x(x^*; r) = r + 3(x^*)^2 = -2r > 0.$$

Thus the nonzero branches are unstable. For $r > 0$ and $x_0 \neq 0$, the cubic term drives solutions away from the origin; the local cubic model contains no large-amplitude saturation.

Question: How does a quintic term saturate a subcritical pitchfork?

Analyze

$$\dot{x} = rx + x^3 - x^5 = x(r + x^2 - x^4).$$

Solution. For a nonzero equilibrium, set

$$y = x^2 > 0.$$

Then

$$y^2 - y - r = 0, \qquad y_\pm = \frac{1 \pm \sqrt{1+4r}}{2}.$$

The two amplitude levels meet at

$$r = -\frac{1}{4}, \qquad y = \frac{1}{2}, \qquad x = \pm\frac{1}{\sqrt{2}}.$$

The equilibrium counts are

$$\begin{array}{c|c} r < -1/4 & x^* = 0 \\ r = -1/4 & x^* = 0,\ \pm 1/\sqrt{2} \\ -1/4 < r < 0 & x^* = 0,\ \pm\sqrt{y_-},\ \pm\sqrt{y_+} \\ r = 0 & x^* = 0,\ \pm 1 \\ r > 0 & x^* = 0,\ \pm\sqrt{y_+}. \end{array}$$

At a nonzero equilibrium,

$$\begin{aligned} f_x(x; r) &= (r + x^2 - x^4) + x(2x - 4x^3) \\ &= 2x^2(1 - 2x^2) = 2y(1 - 2y). \end{aligned}$$

Hence the y_- branches are unstable and the y_+ branches are asymptotically stable. The origin has rate r. At $r = -1/4$,

$$f(x) = -x\left(x^2 - \frac{1}{2}\right)^2,$$

Question: How does a quintic term saturate a subcritical pitchfork? (continued)

so $x = \pm 1/\sqrt{2}$ are half-stable and Lyapunov unstable. At $r = 0$, the origin is unstable and $x = \pm 1$ are asymptotically stable. Therefore

$$-\frac{1}{4} < r < 0$$

has three asymptotically stable equilibria: the origin and two distinct large-amplitude equilibria, one positive and one negative. Suppose r is varied slowly enough that the state follows its current asymptotically stable equilibrium branch until that branch loses stability or ends. As r increases, the origin loses stability at $r = 0$. As r decreases, the large-amplitude branches end at $r = -1/4$.

Question: What bifurcation is encoded by each leading equation?

For $j = 1, 2$, let $R_j(x, \mu)$ be continuous in (x, μ), odd in x, and continuously differentiable in x, with

$$R_j(x, \mu) = O(x^5), \qquad \partial_x R_j(x, \mu) = O(x^4)$$

as $x \to 0$, uniformly for μ near 0. Classify

$$\dot{x} = \mu x - 4x^3 + R_1(x, \mu), \qquad \dot{x} = -\mu x + 2x^3 + R_2(x, \mu).$$

Solution. For the first equation, division by $x \neq 0$ gives

$$0 = \mu - 4x^2 + O(x^4), \qquad x^2 = \frac{\mu}{4} + O(\mu^2).$$

Thus the leading nonzero branches are

$$x_\pm^* = \pm\frac{\sqrt{\mu}}{2} + O(\mu^{3/2}), \qquad \mu > 0, \qquad (\mu \downarrow 0).$$

The leading rates are

$$f_x(0; \mu) = \mu, \qquad f_x(x_\pm^*; \mu) = -2\mu + O(\mu^2) < 0$$

for all sufficiently small $\mu > 0$, with the expansion understood as $\mu \downarrow 0$. It is supercritical.
For the second equation,

$$0 = -\mu + 2x^2 + O(x^4), \qquad x^2 = \frac{\mu}{2} + O(\mu^2),$$

so the leading nonzero branches are

$$x_\pm^* = \pm\sqrt{\frac{\mu}{2}} + O(\mu^{3/2}), \qquad \mu > 0, \qquad (\mu \downarrow 0),$$

with leading rates

$$f_x(0; \mu) = -\mu, \qquad f_x(x_\pm^*; \mu) = 2\mu + O(\mu^2) > 0$$

for all sufficiently small $\mu > 0$, with the expansion understood as $\mu \downarrow 0$. With the standard parameter $R = -\mu$, these unstable nonzero branches exist for $R < 0$, so the pitchfork is subcritical.